Unit 1

Topic 1: Introduction to Artificial Intelligence

Artificial Intelligence (AI) computer science ka field hai jo machines ko **human jaisi intelligence** dene par focus karta hai. AI systems data ko analyze karke decisions le sakte hain. AI ka main goal hai machines ko **learning, reasoning aur problem solving** capable banana. Normal programs fixed rules follow karte hain, but AI systems experience se improve hote hain. AI automation aur smart decision making me use hota hai. AI ka use healthcare, education, banking aur robotics me hota hai.
Example: Google Maps shortest route batata hai aur YouTube recommended videos suggest karta hai. Ye AI ke practical examples hain.

Topic 2: Features of AI

AI ki main feature hai **learning ability**, yani system data se learn karke improve hota hai. AI me **reasoning ability** hoti hai jo logic se decision lene me help karti hai. AI systems **problem solving** me use hote hain jaha complex problems ko step-by-step solve kiya jata hai. AI me **perception** bhi hota hai jisse wo images, voice aur text samajh sakta hai. AI **automation** provide karta hai jisse repetitive work automatically ho jata hai. AI ka main feature hai intelligent decision making.
Example: Face unlock mobile me AI image detect karta hai. Chatbots automatically customer queries solve karte hain.

Topic 3: Agents and Environment

AI me **Agent** ek aisa entity hota hai jo environment ko observe karta hai aur uske according action leta hai. Agent sensors se input leta hai aur actuators se output deta hai. Agent ka main goal hota hai best possible action choose karna. **Environment** wo surrounding hota hai jaha agent kaam karta hai. Agent aur environment ke interaction se AI system work karta hai. Ye concept intelligent behaviour ko design karne me important hai.
Example: Self-driving car me camera sensor hota hai aur steering actuator hota hai. Road aur traffic uska environment hota hai.

Topic 4: Structure of Agents

AI agent ki structure do main parts se milkar banti hai: **Architecture** aur **Agent Program**. Architecture hardware ya platform hota hai jaha agent run karta hai. Agent program wo logic hota hai jo input ke basis par decision leta hai. Agent sensors se data leta hai aur actuators se action perform karta hai. Agent ka behaviour program design par depend karta hai. Simple flow hota hai: **Input Processing Output**.
Example: Robot me camera aur motors architecture hote hain, aur movement decide karne wala software agent program hota hai.

Topic 5: Problem Solving Agents

Problem Solving Agent ek AI agent hota hai jo **goal achieve karne ke liye best solution search karta hai**. Ye agent problem ko states aur actions me divide karta hai. Ye possible solutions explore karta hai aur best path select karta hai. Is process me search algorithms use hote hain. Agent initial state se goal state tak pahunchne ki planning karta hai. Iska focus efficient aur correct solution find karna hota hai.

Example: Google Maps current location se destination tak fastest route find karta hai. Chess AI best move choose karta hai goal win karne ke liye.

Topic 6: Problem Formulation

Problem Formulation ka matlab hai problem ko **clear structure me definarna** taaki AI easily solve kar sake. Isme initial state (start point) define ki jati hai. Goal state (final target) decide kiya jata hai. Possible actions aur states identify kiye jate hain. Path cost bhi consider kiya jata hai jaise time ya distance. Proper formulation se search algorithm easily apply ho sakta hai. Ye AI problem solving ka important step hai.
Example: Route finding me start city initial state hota hai aur destination goal state. Routes possible actions hote hain.

Topic 7: AI Techniques (Search, Knowledge)

AI me problems solve karne ke liye **Search** aur **Knowledge** techniques use hoti hain. Search technique me AI different possible solutions explore karta hai. Ye initial state se goal state tak path find karta hai. Knowledge technique me facts aur rules store kiye jate hain jise **knowledge base** bolte hain. AI decision lene ke liye knowledge aur search dono ka use karta hai. Ye techniques intelligent behaviour design karne me help karti hain.
Example: Spam filter email ke words (knowledge) check karta hai aur decide karta hai spam hai ya nahi (search). Medical AI symptoms ke data se disease suggest karta hai.

Unit 2

Topic 1: Searching for Solutions

Searching for solutions ka matlab hai AI ka **initial state se goal state tak solution find karna**. AI problem ko states aur actions me represent karta hai. Search process me AI different paths explore karta hai. Ye search tree ya graph ka use karta hai. Nodes states ko represent karte hain aur edges actions ko. Goal milne par search stop ho jata hai. Searching AI problem solving ka basic concept hai.
Example: Maze problem me entry initial state hoti hai aur exit goal state. AI different paths try karke correct exit find karta hai.

Topic 2: Breadth First Search (BFS)

Breadth First Search (BFS) ek search algorithm hai jo **level by level nodes explore karta hai**. Ye start node se shuru karke pehle uske nearest nodes visit karta hai. BFS me **Queue (FIFO)** data structure use hota hai. Ye algorithm shortest path find karne me useful hota hai (unweighted graph me). BFS complete search karta hai isliye solution milne ki guarantee hoti hai. But ye memory jyada use karta hai.
Example: Social network me minimum connections find karne ke liye BFS use hota hai. Tree traversal me bhi BFS level order me nodes visit karta hai.

Topic 3: Depth First Search (DFS)

Depth First Search (DFS) ek search algorithm hai jo **pehle depth me jata hai phir backtrack karta hai**. Ye ek path ko end tak explore karta hai before dusra path try kare. DFS me **Stack (LIFO)** ya recursion use hota hai.

Ye memory BFS se kam use karta hai. DFS shortest path guarantee nahi karta. Agar graph deep ho to ye infinite path me ja sakta hai. DFS tab useful hai jab solution depth me ho.
Example: Maze solve karte time ek path ko end tak follow karna DFS approach hai. File system traversal me bhi DFS use hota hai.

Topic 4: Heuristic Search

Heuristic Search ek AI search method hai jo **estimate (heuristic function)** use karke solution fast find karta hai. Ye blind search se smarter hota hai kyunki ye promising paths ko pehle check karta hai. Heuristic function ko **h(n)** se denote karte hain jo goal tak ka estimated cost batata hai. Ye unnecessary paths avoid karta hai. Performance heuristic accuracy par depend karti hai. Galat heuristic se wrong solution mil sakta hai.
Example: Google Maps distance estimate karke best route suggest karta hai. 8-puzzle problem me wrong tiles count heuristic hota hai.

Topic 5: Hill Climbing

Hill Climbing ek **heuristic search algorithm** hai jo current state se better state ki taraf move karta hai. Ye greedy approach follow karta hai yani jo option best lagta hai usko choose karta hai. Ye local optimum par stuck ho sakta hai. Isme backtracking generally nahi hota. Ye simple aur fast algorithm hai. Optimization problems me iska use hota hai.
Example: Agar tum hill par chadh rahe ho to tum hamesha upar ki taraf jaoge. But kabhi small hill par ruk sakte ho even if bigger hill aage ho (local optimum).

Topic 6: A* Algorithm

A* (A star) ek **best-first search algorithm** hai jo search aur heuristic dono use karta hai. Ye **actual cost g(n)** aur **estimated cost h(n)** ko add karke best path find karta hai. Formula hota hai: **f(n) = g(n) + h(n)**. Ye optimal solution de sakta hai agar heuristic correct ho. Ye BFS se faster aur smarter hota hai. Path finding problems me commonly use hota hai.
Example: Google Maps shortest path find karne ke liye A* jaise algorithms use karta hai. Games me characters ko destination tak pahunchane me bhi use hota hai.

Topic 7: AO* Algorithm

AO* Algorithm ek search algorithm hai jo **AND-OR graphs** par work karta hai. Isme problem ko sub-problems me divide kiya jata hai. AND node ka matlab hota hai sab sub-problems solve karne padenge. OR node ka matlab hota hai kisi ek path ko choose karna. Ye heuristic values use karke best solution graph banata hai. AO* problem reduction approach follow karta hai. Ye complex problem solving me useful hai.
Example: Medical diagnosis me disease find karne ke liye multiple tests (AND) aur alternative treatments (OR) ho sakte hain. AO* best combination choose karta hai.

Topic 8: Problem Reduction

Problem Reduction technique me **complex problem ko chote sub-problems me divide** kiya jata hai taaki solution easy ho jaye. Ye divide and conquer approach jaisa concept hai. Har sub-problem ko solve karke final solution banaya jata hai. Isme AND-OR graph ka use hota hai. Ye technique planning aur reasoning problems me useful hoti hai. Complex AI problems ko manageable banane me help karti hai.
Example: Agar project banana hai to usko parts me divide karte hain: design, coding, testing. Sab complete hone par project complete hota hai.

Topic 9: Adversarial Search

Adversarial Search AI me **competitive environments** me use hota hai jaha ek opponent hota hai. Isme AI ko dusre player ke possible moves ka bhi analysis karna padta hai. Ye mostly game playing AI me use hota hai. AI apna best move choose karta hai aur opponent ke best response ko bhi consider karta hai. Ye win ya loss situations predict karta hai. Isme Mini-Max jaise algorithms use hote hain.
Example: Chess game me AI apna move decide karte waqt opponent ke next moves ka bhi calculation karta hai. Tic-Tac-Toe AI bhi same concept use karta hai.

Topic 10: Mini-Max Algorithm

Mini-Max Algorithm adversarial search me use hota hai jaha **do players (Max and Min)** hote hain. Max player apna score maximize karta hai aur Min player opponent ka score minimize karta hai. AI possible moves ka game tree banata hai. Har move ka outcome evaluate kiya jata hai. AI best move choose karta hai assuming opponent bhi best move khelega. Ye turn-based games me important hai.
Example: Chess me AI aisa move choose karta hai jisse uska fayda ho aur opponent ka loss ho. Tic-Tac-Toe AI bhi Mini-Max use karta hai.

Topic 11: Multiplayer Games

Multiplayer games me **do se jyada players** hote hain, isliye decision making complex ho jati hai. AI ko multiple opponents ke moves consider karne padte hain. Isme game tree bada ho jata hai. Strategy banana difficult ho jata hai kyunki har player ka alag goal ho sakta hai. AI ko cooperation aur competition dono handle karna padta hai. Is type ke games me advanced search techniques use hoti hain.
Example: Poker me multiple players hote hain jaha AI ko sabke possible moves predict karne padte hain. Online strategy games me bhi multiplayer AI use hota hai.

Topic 12: Problems in Game Playing

Game Playing AI me kuch challenges hote hain. **Large search space** ek problem hai kyunki possible moves bahut jyada hote hain. **Time limitation** bhi issue hai kyunki AI ko fast decision lena padta hai. **Uncertainty** hoti hai jab opponent unpredictable ho. **Complex rules** bhi problem create karte hain. Kabhi perfect evaluation function banana difficult hota hai. In problems ko solve karne ke liye pruning aur heuristics use kiye jate hain.
Example: Chess me possible moves bahut jyada hote hain isliye sab check karna difficult hai. Poker me opponent ka next move predict karna hard hota hai.

Topic 13: Alpha-Beta Pruning

Alpha-Beta Pruning Mini-Max algorithm ka **optimization technique** hai jo unnecessary nodes ko check karne se bachata hai. Isse search fast ho jata hai. **Alpha** best value hoti hai jo Max player guarantee kar sakta hai. **Beta** best value hoti hai jo Min player guarantee kar sakta hai. Agar kisi branch se better result possible nahi hai to us branch ko **prune (skip)** kar dete hain. Ye same result deta hai jo Mini-Max deta hai but faster.
Example: Chess AI kuch moves ko ignore kar deta hai jab pata chal jata hai ki wo moves useful nahi hain. Isse decision fast milta hai.

Topic 14: Evaluation Functions

Evaluation Function ek function hota hai jo **game state ki quality measure karta hai**. Ye batata hai current position good hai ya bad. Jab complete search possible nahi hota tab AI evaluation function use karta hai. Ye

heuristic value return karta hai. Is value ke basis par AI best move decide karta hai. Game playing AI me ye important role play karta hai. Ye approximate decision making me help karta hai.
Example: Chess me evaluation function pieces ki value (queen = high, pawn = low) ke basis par score calculate karta hai. Jiska score high uski position better.

Unit 3

Topic 1: Knowledge Representation

Knowledge Representation ka matlab hai **information ko aise store karna ki AI usko use karke decision le sake**. AI facts, rules aur relationships ko store karta hai. Ye knowledge base ke form me hota hai. Knowledge ko logical form me represent kiya jata hai. Isse reasoning aur problem solving easy hoti hai. Common methods me semantic networks aur rules aate hain.
Example: Medical AI me symptoms aur diseases ka data store hota hai. Us knowledge ke base par AI disease suggest karta hai.

Topic 2: Bayesian Network

Bayesian Network ek **probabilistic graphical model** hai jo variables ke beech relationships show karta hai. Ye probability use karke uncertainty handle karta hai. Isme nodes variables represent karte hain aur edges dependency show karte hain. Ye Bayes theorem par based hota hai. Ye prediction aur diagnosis problems me use hota hai. Conditional probability ka concept use hota hai.
Example: Agar fever aur cough hai to flu ki probability badh jati hai. Medical diagnosis systems Bayesian Network use karte hain.

Topic 3: Temporal Models

Temporal Models AI me **time-based data ko represent aur analyze** karne ke liye use hote hain. Ye past, present aur future states ke relation ko show karte hain. Time ke saath data kaise change hota hai ye study karte hain. Ye sequence data me useful hote hain. Prediction aur pattern analysis me use hote hain. Speech aur weather prediction me bhi important hain.
Example: Weather forecasting me past temperature data use karke future prediction ki jati hai. Stock market trend analysis me bhi temporal models use hote hain.

Topic 4: Hidden Markov Model (HMM)

Hidden Markov Model ek **probability based model** hai jisme system ki actual state directly visible nahi hoti (hidden hoti hai). Sirf outputs visible hote hain. Ye sequence data analyze karne me use hota hai. Isme states, observations aur transition probabilities hote hain. Ye time dependent processes ko model karta hai. Speech recognition aur pattern recognition me use hota hai.
Example: Speech recognition me actual word (hidden state) directly nahi pata hota, but sound signal (observation) se predict kiya jata hai. Weather prediction me bhi use hota hai.

Topic 5: Kalman Filters

Kalman Filter ek **mathematical algorithm** hai jo noisy data me se accurate estimate find karta hai. Ye prediction aur correction steps use karta hai. Pehle system future state predict karta hai, phir new

measurement se correct karta hai. Ye continuous data tracking me useful hai. Robotics aur navigation systems me use hota hai. Ye real-time estimation ke liye important technique hai.
Example: GPS system moving vehicle ki exact position estimate karne ke liye Kalman Filter use karta hai. Self-driving cars me sensor errors correct karne me bhi use hota hai.

Topic 6: Dynamic Bayesian Network

Dynamic Bayesian Network (DBN) Bayesian Network ka extended form hai jo **time-based probabilistic relationships** show karta hai. Ye variables ke changes ko different time steps par represent karta hai. Ye temporal models aur Bayesian Network ka combination hai. Sequence data analysis me useful hota hai. Prediction aur monitoring systems me use hota hai. Ye uncertainty ke saath time dependent reasoning me help karta hai.
Example: Speech recognition me words ka sequence predict karne ke liye DBN use hota hai. Weather prediction me time ke saath changes analyze karne me bhi use hota hai.

Topic 7: Speech Recognition

Speech Recognition AI ki technique hai jisme **human voice ko text ya commands me convert** kiya jata hai. Ye audio signals ko process karke words identify karta hai. Isme pattern recognition aur machine learning use hota hai. Background noise handle karna ek challenge hota hai. Ye virtual assistants aur voice control systems me use hota hai. Accuracy improve karne ke liye large datasets use hote hain.
Example: Google Assistant aur Alexa voice commands samajhkar action perform karte hain. Voice typing feature bhi speech recognition ka example hai.

Unit 4

Topic 1: Concept of Learning

AI me learning ka matlab hai system ka **experience ya data se performance improve karna**. Machine learning me AI patterns identify karta hai. Learning ke through AI better predictions karta hai. Ye training data use karke rules learn karta hai. Learning supervised, unsupervised ya reinforcement type ki ho sakti hai. AI ka goal hota hai error reduce karna aur accuracy badhana.
Example: Email spam filter past emails se learn karta hai aur future spam detect karta hai. Recommendation systems user behaviour se learn karte hain.

Topic 2: Learning Automation

Learning Automation ka matlab hai AI ka **automatically data se learn karna without manual programming**. Isme machine learning algorithms use hote hain. System training data se patterns identify karta hai. Phir ye patterns future decisions me use hote hain. Isse human effort kam hota hai. Automation large data analysis me useful hai. Ye intelligent systems banane me help karta hai.
Example: Netflix automatically user ke watching history se movies suggest karta hai. Email filters automatically spam identify karte hain.

Topic 3: Genetic Algorithm

Genetic Algorithm ek **search aur optimization technique** hai jo natural selection concept par based hai. Ye population of solutions se start hota hai. Best solutions select hote hain (selection). Phir crossover aur mutation se new solutions bante hain. Weak solutions remove ho jate hain. Ye process tab tak chalta hai jab tak best solution na mil jaye. Optimization problems me ye useful hota hai.
Example: Timetable generation me best schedule find karne ke liye Genetic Algorithm use hota hai. Path optimization problems me bhi use hota hai.

Topic 4: Learning by Induction

Learning by Induction me AI **specific examples se general rules learn karta hai**. Ye data observe karke patterns identify karta hai. Is process ko inductive learning bolte hain. Training data se model banaya jata hai. Ye classification aur prediction problems me use hota hai. Machine learning ka ye basic concept hai. Accuracy data quality par depend karti hai.
Example: Agar AI ko cats ki images dikhayi jaye to wo common features (ears, eyes) learn karke new cat image identify kar sakta hai. Student marks data se result prediction bhi induction ka example hai.

Topic 5: Neural Networks

Neural Networks AI ka model hai jo **human brain ke neurons se inspired** hai. Ye nodes (neurons) aur connections se milkar banta hai. Input layer data leti hai, hidden layer process karti hai aur output layer result deti hai. Ye patterns identify karne me strong hote hain. Deep learning me neural networks ka use hota hai. Image recognition aur prediction tasks me useful hain.
Example: Face recognition systems neural networks use karte hain. Handwritten digit recognition (like ATM cheque reading) bhi iska example hai.

Topic 6: Non-Monotonic Reasoning

Non-Monotonic Reasoning ek reasoning method hai jisme **new information aane par AI apna decision change kar sakta hai**. Ye real life reasoning jaisa hota hai. Normal logic me conclusion fixed hota hai, but isme update ho sakta hai. Ye uncertain environments me useful hai. AI assumptions banata hai but new facts se revise karta hai. Expert systems me iska use hota hai.
Example: AI assume karta hai "birds fly". But agar new info mile ki "penguin bird hai", to AI decision change karega kyunki penguin fly nahi karta.

Topic 7: Probabilistic Reasoning

Probabilistic Reasoning me AI **uncertain situations me probability ke basis par decision leta hai**. Jab exact information available nahi hoti tab ye approach use hoti hai. AI different outcomes ki probability calculate karta hai. Ye Bayesian methods par based hota hai. Prediction aur diagnosis problems me useful hai. Ye risk analysis me bhi use hota hai.
Example: Weather app probability batata hai jaise 70% chance of rain. Medical AI symptoms ke basis par disease ki probability batata hai.

Topic 8: Certainty Factors

Certainty Factors (CF) AI me **belief ya confidence level measure** karne ke liye use hote hain. Ye batate hain ki koi conclusion kitna correct ho sakta hai. CF value generally **-1 se +1** ke beech hoti hai. +1 ka matlab full

certainty aur -1 ka matlab false. Ye expert systems me use hota hai. Ye uncertainty handle karne ka simple method hai. Multiple rules combine karke final certainty calculate hoti hai.
Example: Medical system bol sakta hai disease hone ki certainty 0.7 (70%) hai symptoms ke basis par. Ye doctor ko decision support deta hai.

Topic 9: Fuzzy Logic

Fuzzy Logic ek AI technique hai jo **true ya false ke instead partial truth (0 se 1 ke beech)** use karti hai. Real life me sab decisions binary nahi hote, isliye fuzzy logic useful hai. Ye approximate reasoning allow karta hai. Isme membership values use hoti hain. Control systems me iska use hota hai. Ye uncertainty aur vague data handle karta hai.
Example: AC system me temperature "hot", "warm", "cold" categories me decide hota hai exact number se nahi. Washing machines me water level control bhi fuzzy logic use karta hai.

Unit 5

Topic 1: Robotic Perception

Robotic Perception ka matlab hai robot ka **sensors ke through environment ko samajhna**. Robot cameras, microphones aur other sensors use karta hai. Ye data process karke objects identify karta hai. Computer vision aur pattern recognition ka use hota hai. Ye robots ko intelligent behaviour dene me help karta hai. Navigation aur object detection me important role hai.
Example: Warehouse robots boxes identify karke correct place par rakhte hain. Self-driving cars road signs detect karte hain using cameras.

Topic 2: Localization

Localization ka matlab hai robot ka **apni exact position aur orientation find karna** environment me. Robot sensors jaise GPS, LiDAR aur cameras use karta hai. Ye map ke reference se apni location calculate karta hai. Accurate localization navigation ke liye important hai. Ye robotics aur self-driving vehicles me use hota hai. Errors reduce karne ke liye filters (jaise Kalman filter) use hote hain.
Example: Self-driving car GPS aur sensors se apni position road par detect karti hai. Delivery robot building me apna location track karta hai.

Topic 3: Mapping Configuration Space

Mapping Configuration Space ka matlab hai robot ke **possible positions aur movements ko represent karna**. Isme robot ke size, shape aur obstacles consider kiye jate hain. Ye safe movement planning me help karta hai. Configuration space (C-space) me robot ko ek point ki tarah treat kiya jata hai. Obstacles ko expanded form me represent kiya jata hai. Ye collision avoid karne me useful hai.
Example: Agar robot ko room me move karna hai to table aur chair ko obstacles maan kar safe path plan karta hai. Industrial robots bhi collision avoid karne ke liye use karte hain.

Topic 4: Planning Uncertain Movements

Planning Uncertain Movements ka matlab hai robot ka **uncertain conditions me movement plan karna**. Real world me sensors perfect data nahi dete, isliye uncertainty hoti hai. Robot probability models use karke safe

decisions leta hai. Isme risk aur error ko consider kiya jata hai. AI algorithms best possible action choose karte hain. Ye autonomous robots ke liye important hai.
Example: Self-driving car rain ya fog me slow speed choose karti hai kyunki sensor data uncertain hota hai. Delivery robot obstacles avoid karte hue safe path choose karta hai.

Topic 5: Dynamics and Control

Dynamics and Control robot ke **movement aur stability ko manage karne** se related hai. Dynamics me robot ki speed, force aur motion study ki jati hai. Control systems robot ko correct direction aur speed maintain karne me help karte hain. Feedback sensors robot ko errors correct karne me help karte hain. Ye smooth aur safe movement ke liye important hai. Robotics me PID controllers jaise methods use hote hain.
Example: Drone apni balance maintain karne ke liye continuous speed adjust karta hai. Robot arm correct position par rukne ke liye control system use karta hai.

Topic 6: Ethics and Risks of AI

Ethics and Risks of AI ka matlab hai AI ke **safe aur responsible use** ke baare me sochna. AI se job loss ka risk ho sakta hai due to automation. Privacy issues bhi hote hain kyunki AI large data use karta hai. Wrong decisions ka risk hota hai agar data biased ho. AI misuse bhi ho sakta hai jaise deepfakes. Isliye ethical guidelines aur regulations important hain. Responsible AI development zaroori hai.
Example: Face recognition ka misuse privacy issue create kar sakta hai. Automated hiring system biased data ki wajah se unfair decisions le sakta hai.

Topic 7: Case Study of AI in Robotics

AI in Robotics ka matlab hai robots me **AI techniques use karke intelligent behaviour develop karna**. AI robots ko decision making aur learning ability deta hai. Industrial robots automation ke liye use hote hain. Service robots hospitals aur hotels me use hote hain. AI robots sensors data se environment samajhte hain. Ye efficiency aur accuracy improve karte hain.
Example: Amazon warehouse robots automatically products pick aur move karte hain. Surgical robots doctors ko precise operations me help karte hain.